MEENAKSHI COLLEGE OF ENGINEERING

Since 2001

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT TITLE

AgriSense : A NextGen Weather- Driven Agricultural Intelligence System

TEAM MEMBERS

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PROBLEM STATEMENT

Traditional weather platforms provide broad, non-localized forecasts that fail to meet the specific needs of farmers. Due to the lack of real-time, location-based insights, farmers struggle to make informed decisions regarding crop planning, irrigation, and risk management. Existing systems mostly offer static advisories and do not integrate AI driven intelligence. As a result, weather data is not converted into actionable guidance. This gap leads to poor agricultural outcomes and reduced productivity

ABSTRACT

AgriSense is a smart agriculture intelligence system developed using Machine Learning and real-time weather data. The system analyzes environmental conditions such as temperature, humidity, and rainfall to provide accurate crop recommendations. It also suggests suitable fertilizers and pesticides to improve farming productivity. The XGBoost algorithm is used for efficient prediction and analysis. Real-time weather information is collected through API integration for better decision-making. The system includes features like weather analytics, crop planning, and dashboard insights. AgriSense helps farmers reduce crop loss, improve productivity, and support sustainable farming practices.

INTRODUCTION

- Agriculture highly depends on weather and environmental conditions.
- Farmers face difficulties in crop planning, fertilizer usage, and disease management.
- Traditional farming methods may lead to crop loss and reduced productivity.
- AgriSense provides intelligent agricultural recommendations using Machine Learning and real-time weather data.

LITERATURE SURVEY

S.No	Author	Title	Publication & Year	Pros	Cons
1	Verma, S., and Patel, K.	Forecasting of rainfall using novel SVM and comparison with logistic regression algorithm	Int. J. Advanced Research in Computer Science, 2025	• Provides improved rainfall forecasting accuracy. • Compares SVM with Logistic Regression effectively.	• Requires large climatic datasets. • Prediction accuracy may vary in extreme weather conditions.
2	Mahale, Y., et al.	Crop recommendation and forecasting system using machine learning with LSTM	Sustainability, 2024	• Uses LSTM for better forecasting performance. • Supports intelligent crop recommendation.	• High computational complexity. • Requires continuous training for improved accuracy.
3	Tripathi, P. K., and Mishra, S.	Weather-based crop recommendation system using machine learning	Proc. ICOSEC, 2021	• Provides weather-based crop suggestions. • Helps improve agricultural productivity.	• Limited real-time weather integration. • Recommendations may not support all regions.
4	Pawar, R. B., and Patil, S. B.	Smart agriculture system using IoT and machine learning	IEEE GCWCN, 2021	• Integrates IoT with machine learning. • Enables smart monitoring of agricultural conditions.	• IoT deployment cost is high. • Requires stable internet connectivity.
5	Prasad, S. R. S., et al.	Machine learning techniques for crop yield prediction based on climatic conditions	IEEE ICCIC, 2020	• Improves crop yield prediction accuracy. • Uses climatic conditions for analysis.	• Performance depends on dataset quality. • Limited scalability for large agricultural datasets.

EXISTING SYSTEM

Traditional agricultural systems mainly depend on manual observation and farmer experience for crop planning and management. Existing systems provide only basic weather forecasting and limited agricultural guidance. Most platforms do not offer real-time analysis or intelligent crop recommendations based on environmental conditions. Farmers often face difficulties in selecting suitable crops, fertilizers, and pesticides. Weather predictions are usually generalized and may not accurately represent local farm conditions. Existing systems also lack integrated features for disease management and crop planning. These limitations may lead to reduced productivity, crop loss, and inefficient farming practices. Therefore, a smarter and more efficient agricultural intelligence system is required.

PROPOSED SYSTEM

- A smart farming platform that:
- Uses real-time weather data for localized insights
- Employs XGBoost to recommend crops best suited to current conditions
- Provides a dashboard with analytics, charts, and planning tools
- Converts weather parameters into meaningful, actionable guidance
- Enhances agricultural outcomes through AI + real-time intelligence

SOFTWARE AND HARDWARE REQUIRMENTS

SOFTWARE REQUIREMENTS:

1.**Frontend:** HTML, TypeScript, Tailwind CSSp
2.**Backend**: Python
3.**ML Tools:** XGBoost
4.**API:** OpenWeatherMap
5.**Development Tools:** VS Code

HARDWARE REQUIREMENTS:

1. **Processor:** Intel Core i5 or above
2. **RAM:** Minimum 8 GB (16 GB recommended for model processing)
3. **Storage:** At least 500 GB HDD or 256 GB SSD
4. **Graphics:** Integrated GPU (Dedicated GPU recommended for faster processing)
5. **Internet:** Stable connection for real-time weather API access

LITERATURE GAP ANALYSIS

IDENTIFIED GAPS IN EXISTING RESEARCH:

1) Most existing systems provide only basic weather forecasting without intelligent agricultural recommendations.

2) Many crop prediction systems lack real-time weather data integration and location-specific analysis.

3) Existing research mainly focuses on crop prediction and does not fully integrate fertilizer and pesticide recommendations.

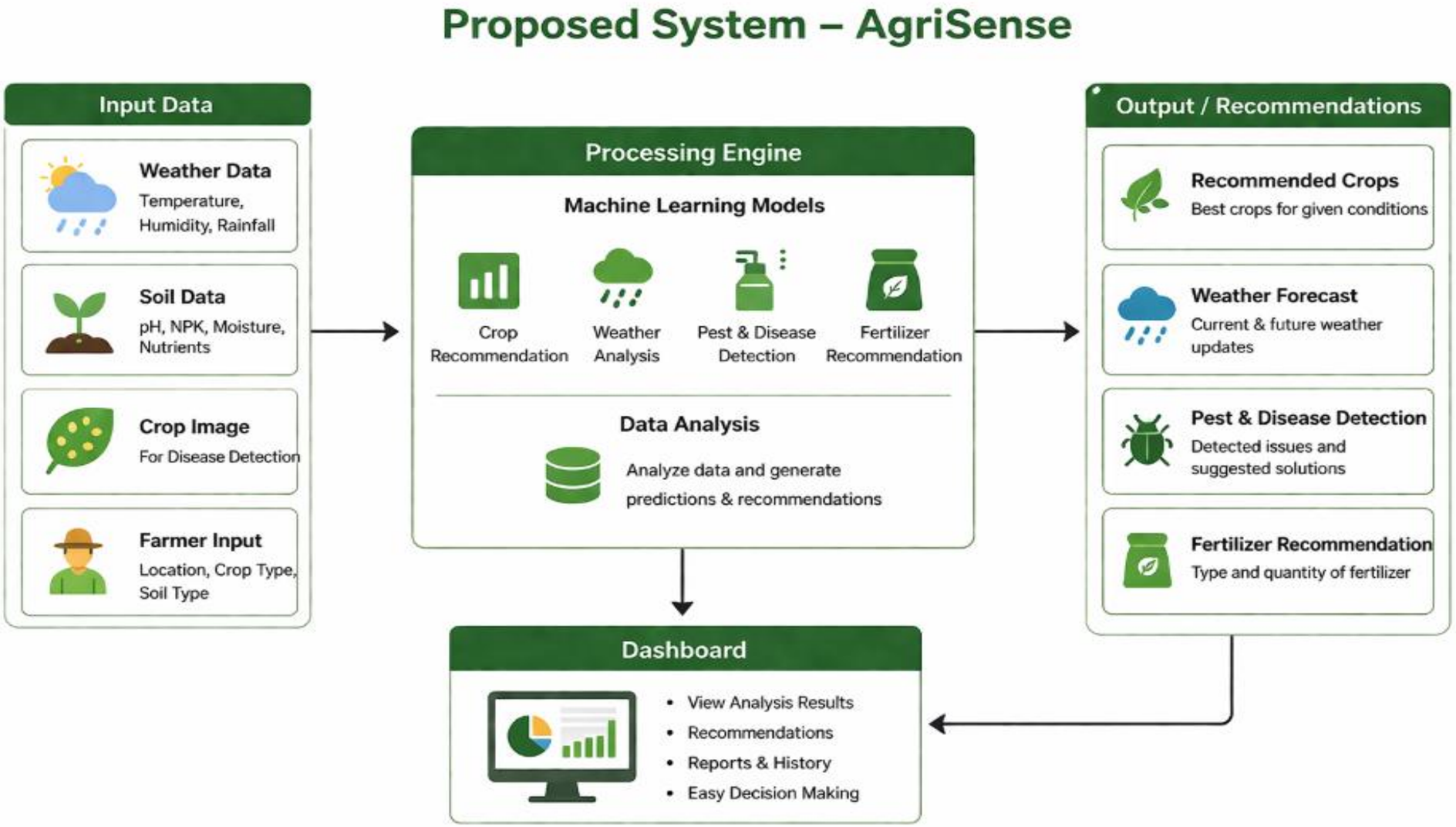
HOW OUR PROJECT BRIDGES THESE GAPS:

1) AgriSense integrates real-time weather data with machine learning for accurate agricultural recommendations.

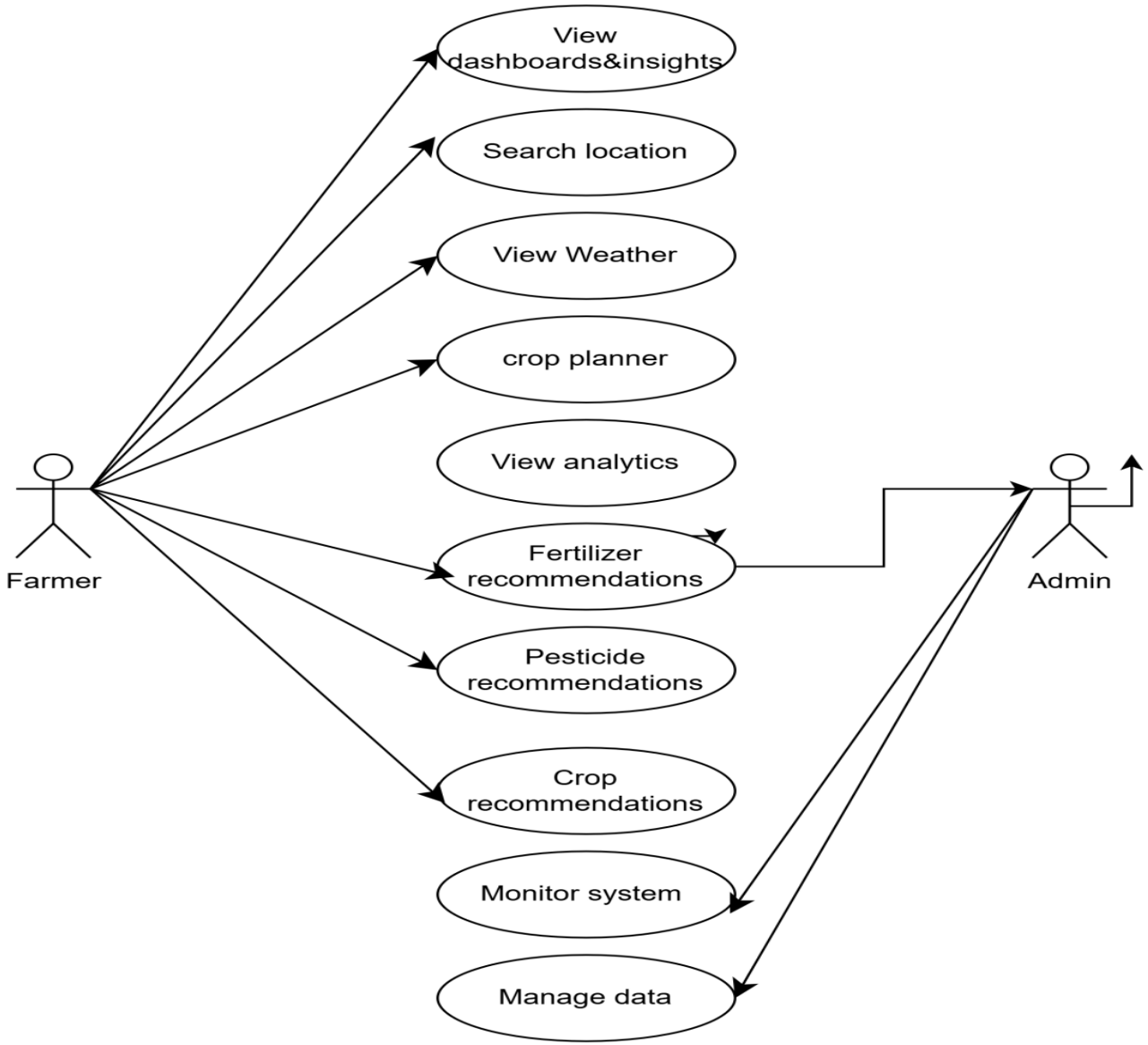
2) The system uses the XGBoost algorithm to provide better prediction accuracy and location-based analysis.

3) AgriSense combines crop recommendation, fertilizer suggestion, pesticide guidance, and weather analytics into a single user-friendly platform

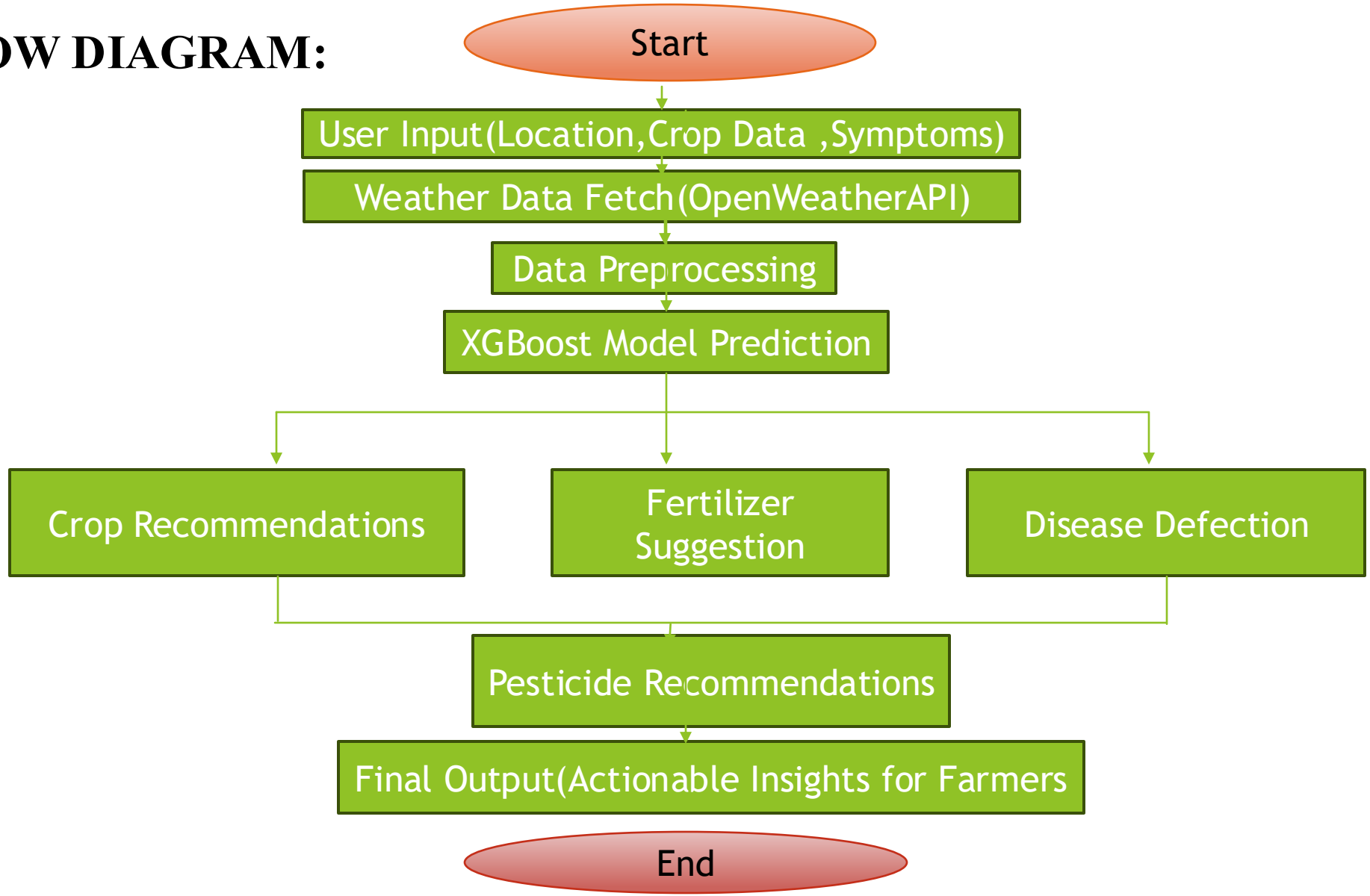
ARCHITECTURE



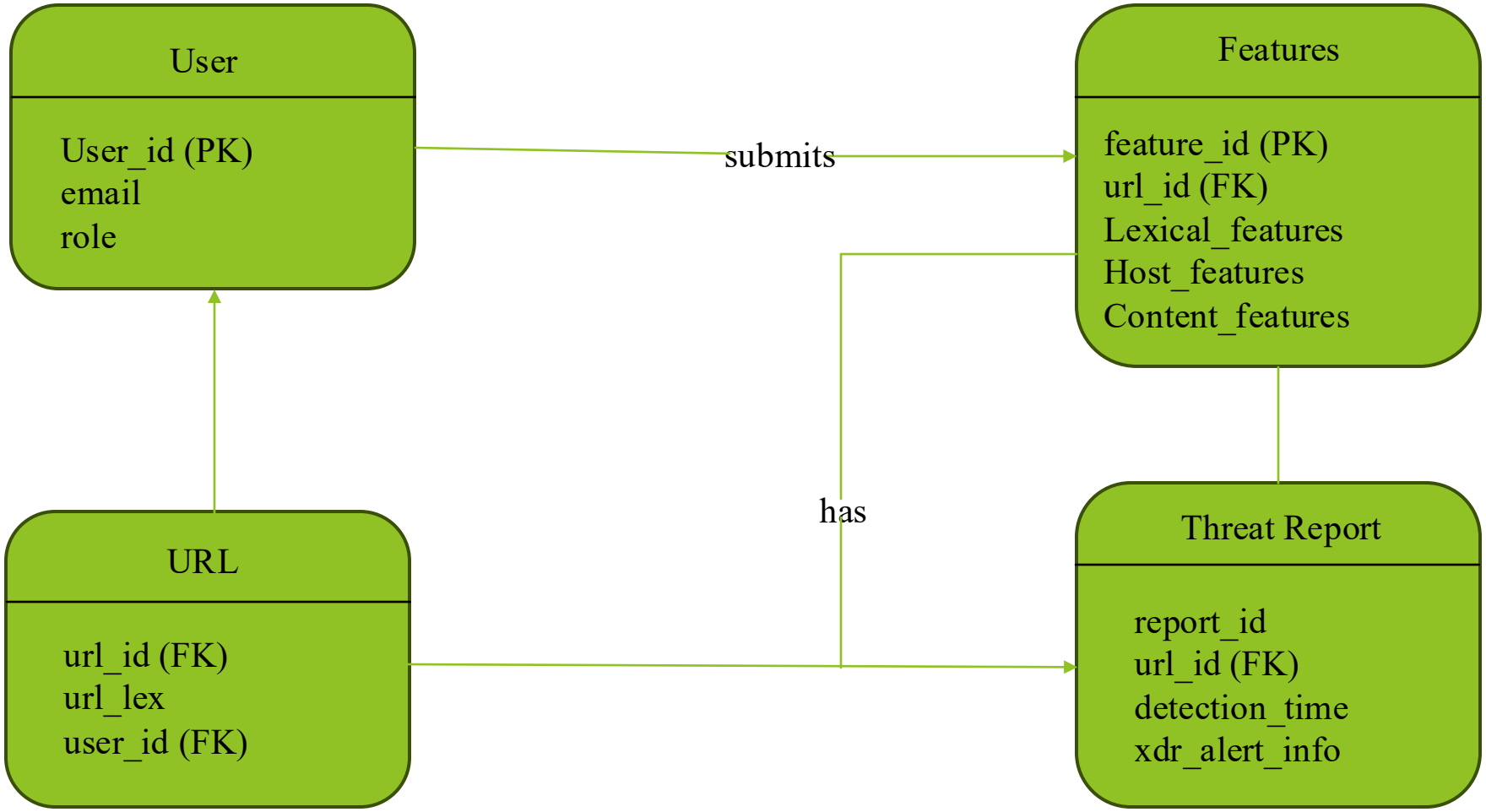
USE CASE DIAGRAM



DATA FLOW DIAGRAM:



ENTITY RELATIONSHIP DIAGRAM



LIST OF MODULES

1) Weather Dashboard Module
2) Weather Analytics Module
3) Crop Recommendation Module
4) API Integration Module
5) Crop Planner Module
6) Fertilizer Recommendation Module
7) Pesticide Recommendation Module

MODULE 1 : WEATHER DASHBOARD MODULE

1) Displays real-time weather data such as temperature, humidity, and rainfall
2) Provides a clear overview of current environmental conditions
3) Helps users quickly understand weather status
4) Updates data dynamically using API

MODULE 2 : WEATHER ANALYTICS MODULE

1) Analyzes weather data trends over time
2) Displays charts and graphs for better understanding
3) Provides insights like rainfall patterns and temperature variations
4) Helps in predicting risks and planning activities

MODULE 3 : CROP RECOMMENDATION MODULE

1) Uses XGBoost algorithm for prediction
2) Suggests suitable crops based on weather conditions
3) Provides accurate and data-driven recommendations
4) Improves crop selection and productivity

MODULE 4 : CROP PLANNER MODULE

1) Helps users plan crop schedules and activities
2) Provides guidance on planting and harvesting time
3) Supports better farm management decisions
4) Works based on recommended crops and weather data

MODULE 5 : FERTILIZER RECOMMENDATION MODULE

1) Suggests fertilizers based on crop type and growth stage
2) Provides quantity and application timing
3) Improves soil fertility and crop yield
4) Helps in efficient resource usage

MODULE 6 : PESTICIDE RECOMMENDATION MODULE

1) Identifies crop diseases and pest issues
2) Suggests appropriate pesticide solutions
3) Categorizes severity levels (Low, Medium, High)
4) Helps reduce crop damage and losses

CODING

App.tsx:

```
import { QueryClient, QueryClientProvider } from "@tanstack/react-
query";
import { BrowserRouter, Route, Routes } from "react-router-dom";
import { Toaster as Sonner } from "@/components/ui/sonner";
import { TooltipProvider } from "@/components/ui/tooltip";
import Index from "./pages/Index";
import Graphs from "./pages/analytics";
import Recommendations from "./pages/cropplanner";
import NotFound from "./pages/NotFound";
import Fertilizers from "./pages/Fertilizers";
import Pesticides from "./pages/Pesticides";
import Scanner from "@/components/Scanner";
import { Toaster } from "sonner";
import { LocationProvider } from "./context/LocationContext";
```

```
import { HashRouter } from "react-router-dom";
const queryClient = new QueryClient();
 const App = () => (
<QueryClientProvider client={queryClient}>
 <LocationProvider>
<HashRouter>
 <TooltipProvider>
 <Toaster />
 <Routes>
<Route path="/" element={<Index />} />
<Route path="/scanner" element={<Scanner />} />
<Route path="/graphs" element={<Graphs />} />
<Route path="/recommendations" element={<Recommendations />} />
<Route path="/pesticides" element={<Pesticides/>} />
<Route path="/fertilizers" element={<Fertilizers />} />
```

```
<Route path="*" element={<NotFound />} />
 </Routes>
 </TooltipProvider>
 </HashRouter>
 </LocationProvider>
</QueryClientProvider>
);
```

Layout.tsx:

```
import { Link, useLocation } from "react-router-dom";
import { Cloud, BarChart3, Sprout, Home, Camera } from "lucide-react";
import logo from "@/assets/logo.png";
import heroFarm from "@/assets/hero-farm.jpg";
import { FlaskConical } from "lucide-react";
import { Bug } from "lucide-react"; // or use FlaskConical again if you want
// import ChatBot from "@/components/ChatBot";
```

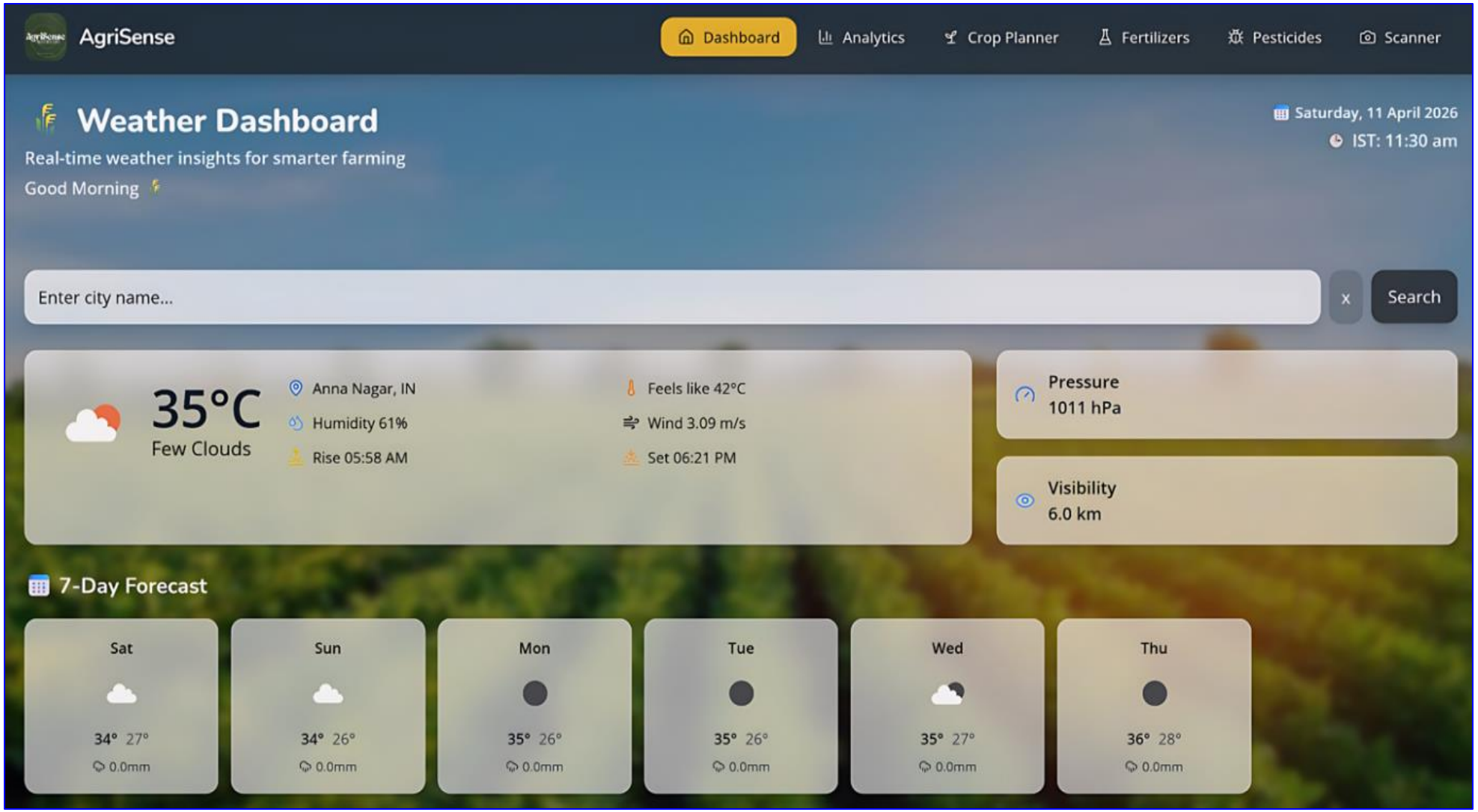
```
const navItems = [
  { path: "/", label: "Dashboard", icon: Home },
  { path: "/graphs", label: "Analytics", icon: BarChart3 },
  { path: "/recommendations", label: "Crop Planner", icon: Sprout },
  { path: "/fertilizers", label: "Fertilizers", icon: FlaskConical },
{ path: "/pesticides", label: "Pesticides", icon: Bug },
{ label: "Scanner", path: "/scanner", icon: Camera },
];
 export default function Layout({ children }: { children: React.ReactNode }) {
const { pathname } = useLocation();
 return (
<div className="relative min-h-screen flex flex-col">
 {/* 🔷 Background Image */}
<divclassName="absolute inset-0 bg-cover bg-center blur-sm scale-105"
style={{ backgroundImage: `url(${heroFarm})` }}
/>
```

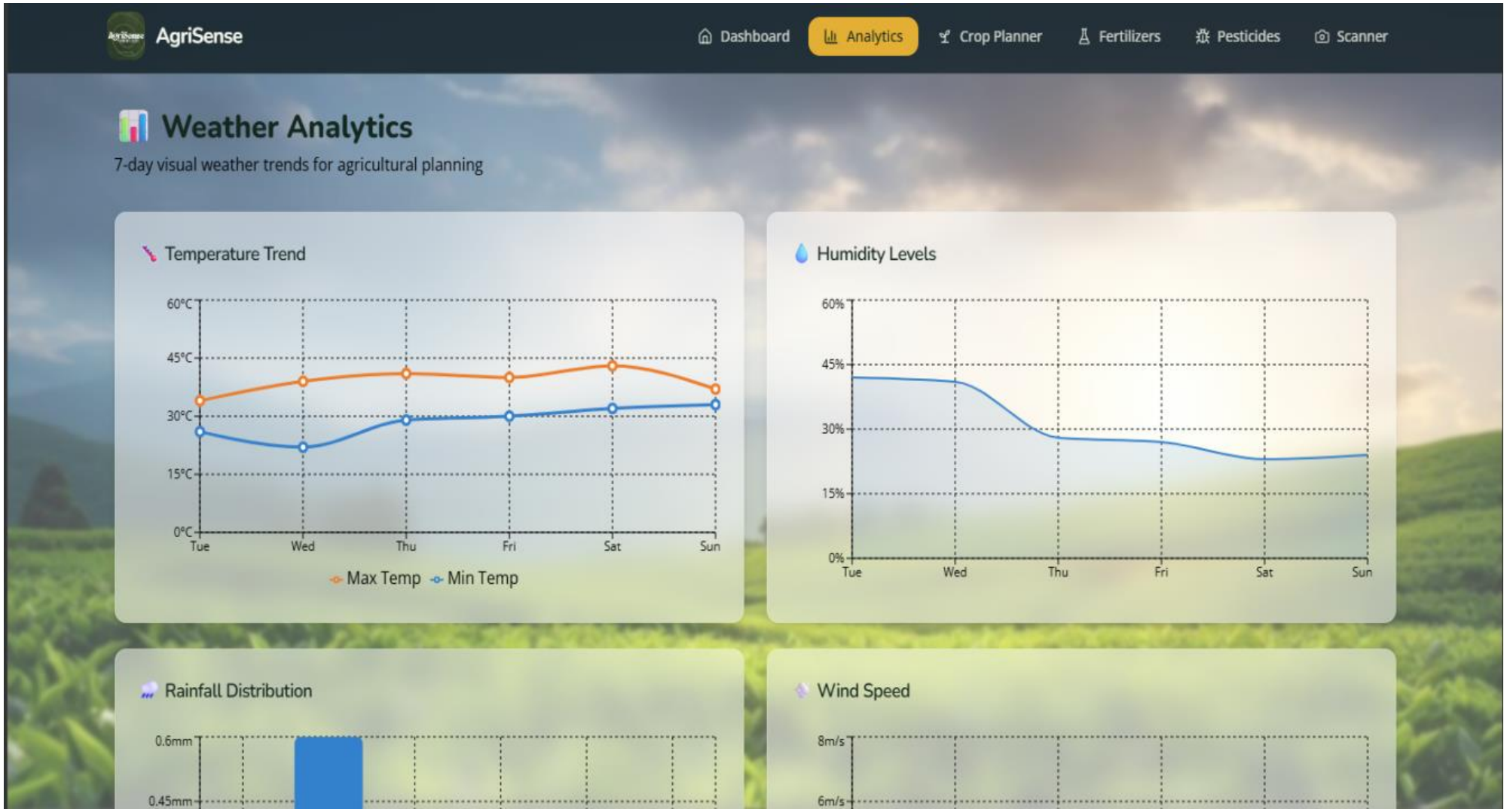
```
{/* 🔷 Overlay for readability */}
<div className="absolute inset-0 bg-black/40" />
 {/* 🔷 Content */}
<div className="relative flex-1 flex-col min-h-screen">
<header className="bg-neutral-900/70 backdrop-blur-lg border-b
border-white/20 shadow-lg  text-center text-sm text-white">
{/* className="bg-neutral-900/70 backdrop-blur-lg border-b border-
white/20 shadow-lg"> */}
<div className="container mx-auto px-4 py-3 flex items-center
justify-between">
<Link to="/" className="flex items-center gap-3">
<img src={logo} alt="WeatherYield" width={40} height={40}
className="rounded-lg" />
<span className="text-xl font-heading font-bold text-primary-
foreground tracking-tight">
AgriSense
```

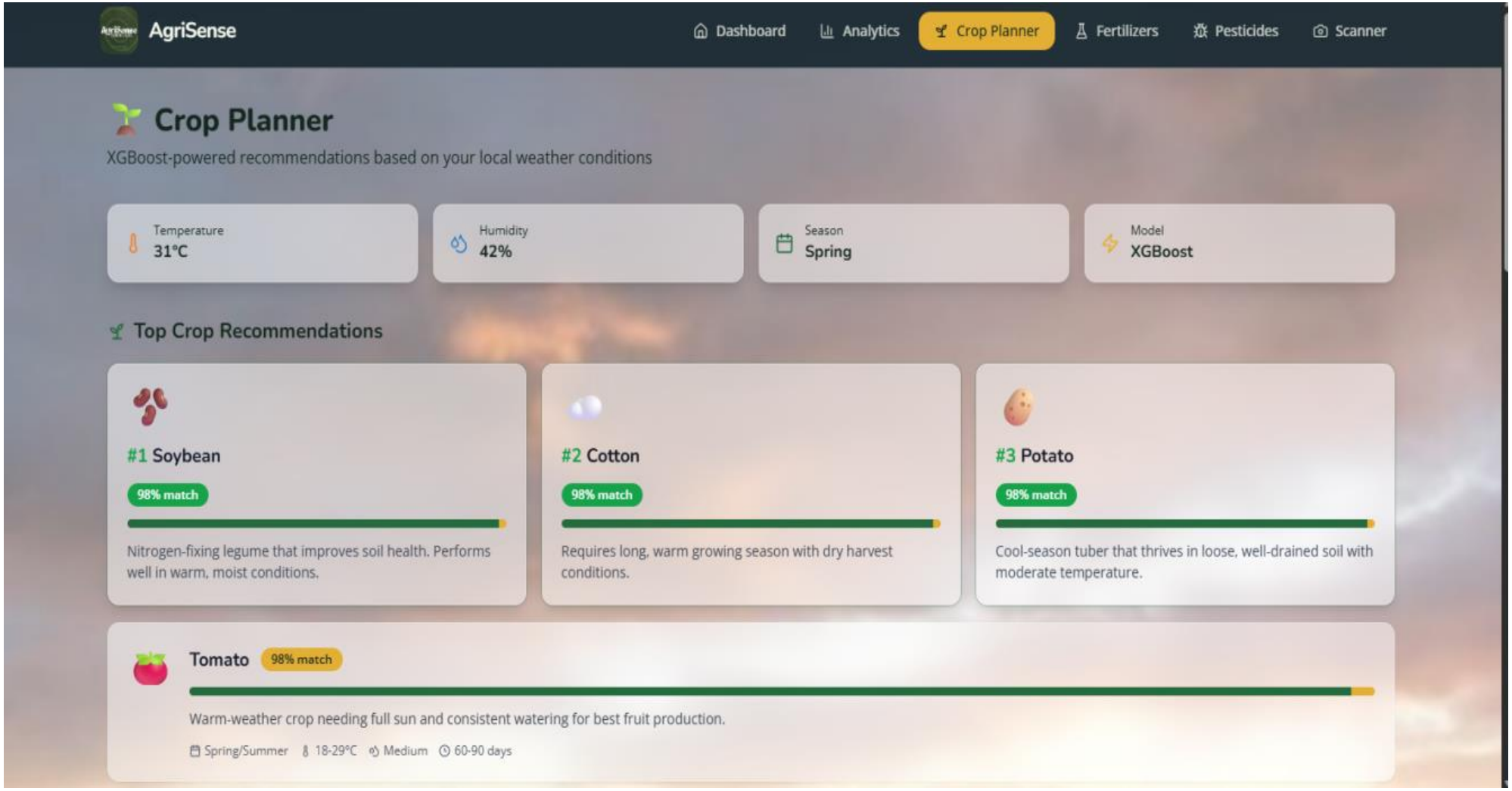
```
  </span>
</Link>
<nav className="flex items-center gap-1">
{navItems.map(({ path, label, icon: Icon }) => (
<Link
key={path}
to={path}
className={`flex items-center gap-2 px-4 py-2 rounded-lg text-sm font-
medium transition-all ${
pathname === path
? "bg-secondary text-secondary-foreground shadow-sm"
```

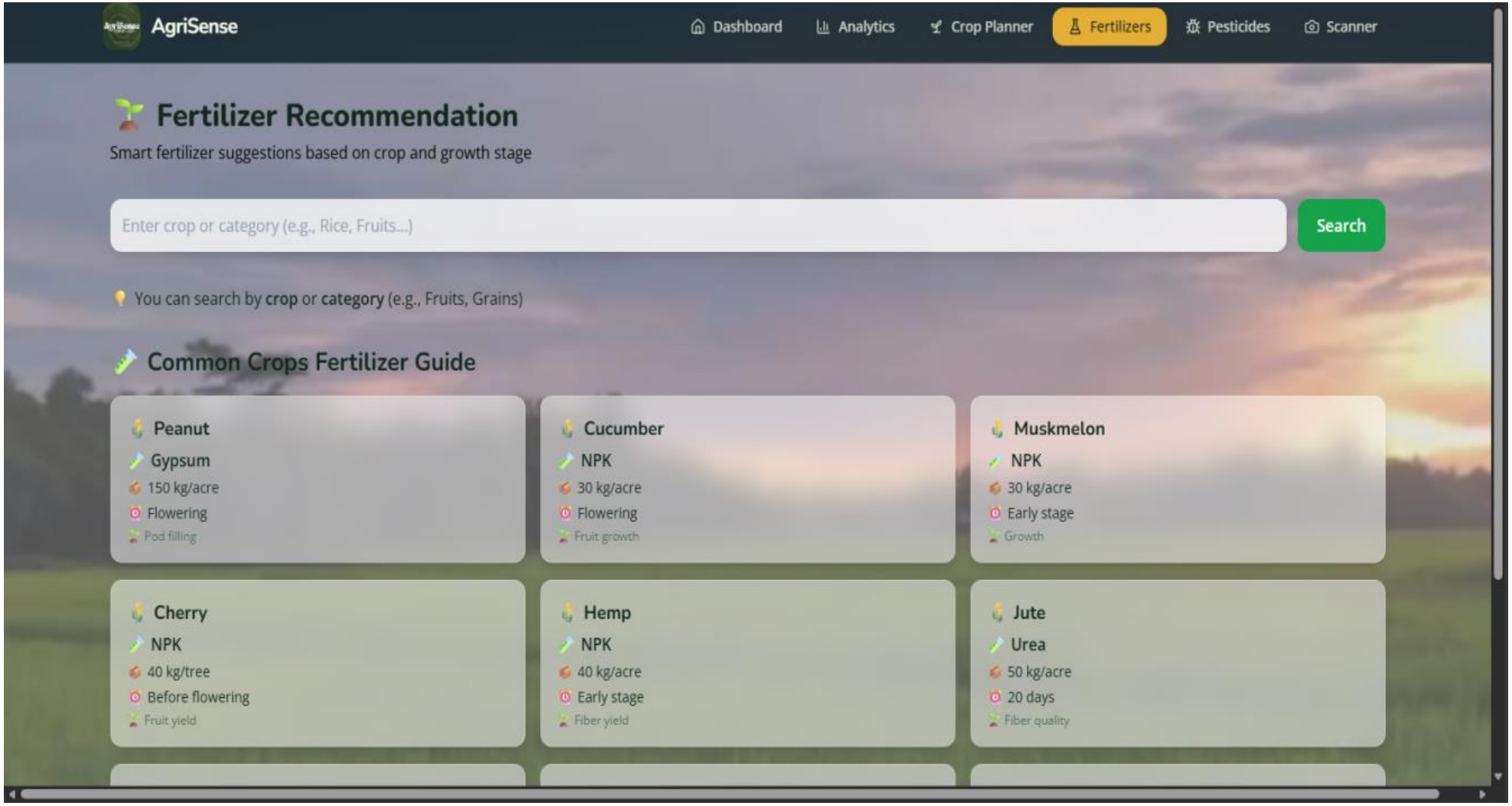
```
: "text-primary-foreground/80 hover:text-primary-foreground hover:bg-primary-foreground/10"
}
</Link>
))}
</nav>
</div>
</header>
<main className="flex-1 flex-col min-h-screen">{children}</main>
{/* <ChatBot /> */}
<footer className="bg-neutral-900/50 backdrop-blur-lg border-b border-white/20 shadow-lg border-t
border-white/20 bg-neutral-900/80 backdrop-blur-md py-4 text-center text-sm text-white">
<div className="container mx-auto px-4 flex items-center justify-center gap-2">
<Cloud className="w-4 h-4" />
<span>AgriSense: A Next-Gen Weather-Driven Agricultural Intelligence System</span>
</div>
</footer>
</div>
</div>
);
}
      }
```

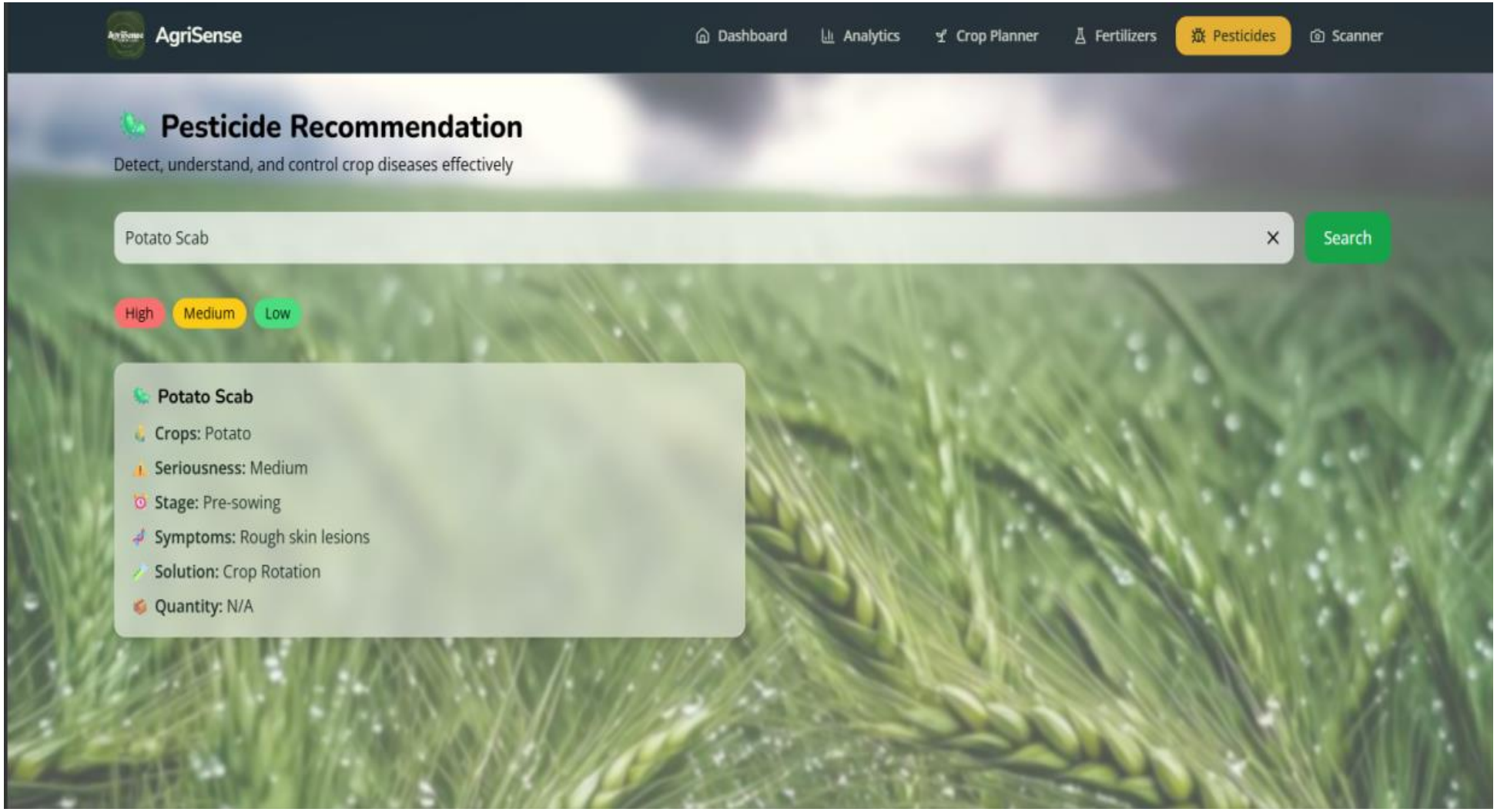
OUTPUT


AgriSense
Dashboard
Analytics
Crop Planner
Fertilizers
Pesticides
Scanner
Weather Dashboard
Real-time weather insights for smarter farming
Good Morning
Saturday, 11 April 2026
IST: 11:30 am
Enter city name...
x
Search
35°C
Few Clouds
Anna Nagar, IN
Humidity 61%
Rise 05:58 AM
Feels like 42°C
Wind 3.09 m/s
Set 06:21 PM
Pressure
1011 hPa
Visibility
6.0 km
7-Day Forecast
Sat
34° 27°
0.0mm
Sun
34° 26°
0.0mm
Mon
35° 26°
0.0mm
Tue
35° 26°
0.0mm
Wed
35° 27°
0.0mm
Thu
36° 28°
0.0mm


AgriSense
Dashboard
Analytics
Crop Planner
Fertilizers
Pesticides
Scanner
Weather Analytics
7-day visual weather trends for agricultural planning
Temperature Trend
60°C
45°C
30°C
15°C
0°C
Tue
Wed
Thu
Fri
Sat
Sun
Max Temp
Min Temp
Humidity Levels
60%
45%
30%
15%
0%
Tue
Wed
Thu
Fri
Sat
Sun
Rainfall Distribution
0.6mm
0.45mm
Wind Speed
8m/s
6m/s


AgriSense
Dashboard
Analytics
Crop Planner
Fertilizers
Pesticides
Scanner
Crop Planner
XGBoost-powered recommendations based on your local weather conditions
Temperature
31°C
Humidity
42%
Season
Spring
Model
XGBoost
Top Crop Recommendations
#1 Soybean
98% match
Nitrogen-fixing legume that improves soil health. Performs well in warm, moist conditions.
#2 Cotton
98% match
Requires long, warm growing season with dry harvest conditions.
#3 Potato
98% match
Cool-season tuber that thrives in loose, well-drained soil with moderate temperature.
Tomato
98% match
Warm-weather crop needing full sun and consistent watering for best fruit production.
Spring/Summer
18-29°C
Medium
60-90 days


AgriSense
Dashboard
Analytics
Crop Planner
Fertilizers
Pesticides
Scanner
Fertilizer Recommendation
Smart fertilizer suggestions based on crop and growth stage
Enter crop or category (e.g., Rice, Fruits...)
Search
You can search by crop or category (e.g., Fruits, Grains)
Common Crops Fertilizer Guide
Peanut
Gypsum
150 kg/acre
Flowering
Pod filling
Cucumber
NPK
30 kg/acre
Flowering
Fruit growth
Muskmelon
NPK
30 kg/acre
Early stage
Growth
Cherry
NPK
40 kg/tree
Before flowering
Fruit yield
Hemp
NPK
40 kg/acre
Early stage
Fiber yield
Jute
Urea
50 kg/acre
20 days
Fiber quality


AgriSense
Dashboard
Analytics
Crop Planner
Fertilizers
Pesticides
Scanner
Pesticide Recommendation
Detect, understand, and control crop diseases effectively
Potato Scab
Search
High
Medium
Low
Potato Scab
Crops: Potato
Seriousness: Medium
Stage: Pre-sowing
Symptoms: Rough skin lesions
Solution: Crop Rotation
Quantity: N/A

FUTURE ENHANCEMENTS

1. Mobile App, Multilingual & Voice Support
2. IoT-Based Real-Time Monitoring
3. Satellite & Large-Scale Data Integration
4. Advanced AI for Irrigation, Yield Prediction & Disease Detection
5. Crop Disease Scanner with Image-Based Detection
6. Cloud Deployment & System Scalability

REFERENCES

1) Verma, S., and Patel, K. (2025), ‘Forecasting of rainfall using novel SVM and comparison with logistic regression algorithm’, *Int. J. Advanced Research in Computer Science*.
2) Mahale, Y., et al. (2024), ‘Crop recommendation and forecasting system using machine learning with LSTM’, *Sustainability*.
3) Tripathi, P. K., and Mishra, S. (2021), ‘Weather-based crop recommendation system using machine learning’, *Proc. ICOSEC*.
4) Pawar, R. B., and Patil, S. B. (2021), ‘Smart agriculture system using IoT and machine learning’, *IEEE GCWCN*.
5) Prasad, S. R. S., et al. (2020), ‘Machine learning techniques for crop yield prediction based on climatic conditions’, *IEEE ICCIC*.

THANK YOU